

# The Self as Integrated Whole Holding Multiple Aspects: A Foundational Architectural Commitment in the Coordination Knowledge Substrate Pattern Extended to AI Selves

**Author:** Wenxin Li (Independent Researcher)

**ORCID:** 0009-0004-8065-3235

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## Attribution

This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize the **Self level** — the integrated whole that holds multiple aspects as facets of one CKS-governed AI intelligence — as a standalone foundational architectural commitment, with operational content distinct from the cell and aspect levels formalized in companion notes.

## Abstract

Paper 2 introduces three architectural levels — cell, aspect, Self — within one CKS-governed AI Self under unified human governance. The Self is the topmost level: the integrated whole that holds multiple aspects as facets of one CKS-governed intelligence. The Self is commonly misread as either a single-system entity (one model is the whole system) or as a rigid top-of-hierarchy that commands subordinate aspects. Both misreadings miss the architectural substance. This note formalizes the Self level as standalone foundational commitment: an integrated whole holding multiple aspects without strict subordination, with cross-level access where purpose requires, and the level at which Paper 2's unifying move — the instinct/reasoning separation — operates as unifying architectural commitment for the entire Self. The note states the commitment precisely, distinguishes it from two adjacent treatments (single-system, rigid-hierarchical), names the not-bigger-aspect distinction as foundational, articulates the cognitive analog (person-with-multiple-modes) as conceptual scaffold, shows how Paper 1's commitments inherit at Self scope, articulates operational implications, names limits, and provides an operational test.

## 1. Why the Self level needs to be formalized as standalone foundational commitment

Paper 2 names three architectural levels — cell, aspect, Self — within one CKS-governed AI Self. A companion note formalizes the cell as atomic unit (B1.03). A second formalizes the aspect as coordination arrangement of cells serving a particular purpose (B1.04). This note formalizes the third level: the Self.

The Self level needs to be named as a standalone architectural commitment for three reasons.

First, the Self is the level at which Paper 2's unifying move operates. The instinct/reasoning separation per B1.01 is at Self scope: the Self has an instinct layer (the LLM) and a reasoning layer (the CKS substrate) composed into one Self under unified human governance. Cells participate in the separation as instances;

aspects coordinate cells; the Self is where the separation is the *unifying* architectural commitment. Without formalizing the Self, Paper 2's central architectural move has no level to land on.

Second, the Self is the unit of deployment. A deployment names, governs, audits, and migrates Selves as wholes — not aspects independently, not cells independently, but Selves as integrated CKS-governed AI intelligences. Subsequent operational notes that refer to "the Self" as a defined object depend on this note for the reference's precision.

Third, the Self is commonly misread. Two adjacent framings dominate conventional AI architecture: the single-system framing (one model is the whole system) and the rigid-hierarchical framing (orchestrator above, subordinate tools below). Both are useful patterns in their own contexts. Neither is the CKS Self. Naming the Self level explicitly is what prevents the slide into either misreading.

## 2. The Self level, defined precisely

In the CKS pattern as extended by Paper 2, the **Self** is the integrated whole that holds multiple aspects as facets of one CKS-governed AI intelligence under unified human governance. The commitment has five operational components.

**(a) Integrated whole, not arbitrary aggregate of aspects.** The Self is not "all the aspects together as a sum." It is the integrated whole that holds aspects as facets. Integration is architectural: substrate-resident content shared across aspects where purpose requires, coherent governance applied under one authority architecture, unified accountability at Self scope. A structure that contains aspects but does not integrate them — independent agents grouped only by deployment configuration — is not a Self in the CKS sense.

**(b) Holds multiple aspects as facets, not as subordinates.** Aspects within the Self are not commanded by the Self in the sense of being subordinate executors. Each aspect retains its own purpose, content domain, and orchestration. The Self is what it is *because* it contains multiple coexisting purposes as one entity, not because it directs them.

**(c) Cross-level access where purpose requires.** Access patterns within the Self are not strictly hierarchical. The Self can access cells directly without requiring aspect mediation when purpose requires. The flexibility is architectural commitment, not implementation detail. A structure that locks all cell access behind aspect-level coordination fails this component.

**(d) The level at which the instinct/reasoning separation operates.** Per B1.01, Paper 2's unifying architectural move — the separation of instinct (LLM, fast pattern-matched response) and reasoning (substrate, deliberate human-governed processing) into independently evolving layers — operates at Self scope. The Self holds both layers as one Self; the separation is the unifying commitment for the integrated whole.

**(e) Operationally addressable as one entity.** The Self is the unit deployments name, govern, and operate over. Self-level operations are first-class architectural objects, not aggregates of lower-level operations.

The five components together define what the Self is, beyond simply "the topmost of three structural levels."

## 3. What makes the Self level architecturally distinctive

Two adjacent framings dominate conventional AI architecture. Each is well-defended in its own context. Each, if confused with the CKS Self, produces a different system.

**The single-system framing.** In conventional LLM-only AI architectures, one model is the whole system. The system has a single inference endpoint, a single weights file, a single locus of behavior. There is nothing for the model to "integrate"; the model is what it does. This framing is well-defended for what it is — a single-model AI system. It is not the CKS Self. The CKS Self holds multiple aspects, has substrate content outside the model that the model does not own, and has coordination logic outside the model that the model does not produce. Calling such a structure a "single system" loses the architectural distinction between the LLM (instinct layer per B1.01) and the substrate (reasoning layer per B1.01) that defines the Self.

**The rigid-hierarchical framing.** A common alternative imagines an orchestrator (often an LLM-driven planner) at the top, calling subordinate tools or agents below. The orchestrator decides; the tools execute. This is a real architectural pattern with real defenders. It is also not the CKS Self. The Self does not "decide" in this command sense; aspects do, within their purposes. The Self does not command its aspects; it contains them. Aspects within the Self are coexisting perspectives, not subordinate executors of Self-level intent. The Self can reach a cell directly when purpose requires, but this is not orchestrator-style command — it is the substrate-resident content's accessibility from the integrated-whole scope under appropriate authority.

The Self is neither of these. It is the integrated whole holding multiple coexisting perspectives without strict subordination, with cross-level access where purpose requires, and where the instinct/reasoning separation operates as unifying architectural commitment.

The **not-bigger-aspect** distinction is foundational. An aspect has a specific purpose and asks pattern questions across its constituent cells *for that purpose*. The Self does not have one purpose; it holds multiple aspects whose purposes coexist. Treating the Self as "the biggest aspect" — an aspect with broader scope — collapses the integrated-whole property into a purpose-defined property and loses the multi-purpose coexistence the Self is the level *for*.

The **not-rigid-hierarchy** distinction is equally foundational. Cross-level access is architectural, not exceptional. The Self reaches cells directly when purpose requires; it does not have to route through aspect mediation. A subsequent Phase B1 note formalizes this access property as standalone foundational commitment.

## 4. The cognitive analog

Paper 2 uses a cognitive analog as conceptual scaffold. A person is not "competitive sports mode" plus "calm study mode" as separate entities glued together; the person is one self who holds these modes as facets. Different modes activate in different contexts. Some capacities (general physical conditioning, language) span the modes; others (the specific repertoire of plays, the studious silence routine) are mode-specific. The modes coexist in one person because that person *is* the integrated whole that holds them. The analog maps directly: aspects are like modes; the Self is like the person; the same underlying CKS artifacts (cells) can participate in multiple aspects, just as a person's general cognition participates in multiple modes.

The architectural substance is not the analog. The analog gives a reader an immediate intuitive grasp of what "integrated whole holding multiple aspects" means. The substance is the integrated-whole property as defined in §2 — substrate-resident content, coherent governance, unified accountability, cross-level access,

the instinct/reasoning separation as unifying commitment. The analog is conceptual scaffold the reader absorbs quickly; the substance is the architecture's own.

## 5. Paper 1 commitments at Self scope

The Self level inherits Paper 1's architectural commitments at the integrated-whole scope. Paper 2's continuity-with-Paper-1 statement is explicit: all Paper 1 architectural commitments hold at every level of Paper 2's architecture. Each carries a specific shape at Self scope worth naming briefly.

**A1.01 (human-governed: authority not labor).** Humans hold the rights to inspect, modify, and override at Self scope. A human with appropriate Self-level authority can inspect substrate content across aspects, modify orchestration rules whose scope is the Self, and override Self-level operations including those that span multiple aspects.

**A1.13 (composition requirements).** A1.13's constraints apply to inter-aspect composition within the Self. When two aspects share cells, share substrate content, or interact through governed orchestration, the composition must satisfy A1.13 at Self scope.

**A1.16 (hybrid systems composition).** A1.16 holds at Self scope, and at Self scope it is the instinct/reasoning separation per B1.01. The Self has an instinct layer (LLM) and a reasoning layer (CKS substrate) composed under unified human governance. The hybrid composition pattern from A1.16 — substrate-mediated, governance-coherent, conflict-aware — is what the instinct/reasoning separation instantiates at Self scope.

**A1.05 (tool-agnosticism).** The Self is tool-agnostic at Self scope. A Self can migrate from one substrate platform to another without losing its identity, provided the new host satisfies A1.05's three minimal requirements (persistent state, human read/write, LLM access).

**A1.07 (path retraceability).** A1.07's provenance metadata applies to Self-level operations including cross-aspect operations. An operation that touches multiple aspects within the Self is retraceable across those aspects through the substrate's provenance metadata.

**A1.10 (the determinism contract).** A1.10's guarantees hold at Self scope. Substrate state at Self scope is deterministic in the senses A1.10 commits to; the allowed non-determinism (LLM inference variability under bounded conditions) holds at Self scope as it does at cell scope.

The inheritance is not a re-derivation. Naming each Paper 1 commitment at Self scope is what makes the Self level a complete architectural commitment rather than a residual top of a stack.

## 6. Operational implications

Several implications follow from formalizing the Self as integrated whole.

**The Self is the unit of deployment.** A deployment names, configures, and operates over Selves. Multiple aspects within one Self are facets of one deployable entity, not separate deployments.

**Self-level governance is a real operational object.** Governance per Paper 1's authority architecture applies at Self scope: humans author orchestration rules whose scope is the Self, exercise override at Self scope, audit operations across aspects within the Self. Self-level governance is not an aggregate of aspect-level governances; it is a distinct authority architecture with the integrated whole as its scope.

**Self-level audit reaches across aspects.** A1.07's provenance metadata makes Self-level audit possible across operations that span multiple aspects. An auditor with Self-level inspect authority can read substrate content across the Self, including content that passes through cross-aspect coordination.

**Self-level migration is the unit of vendor-portability.** Tool-agnosticism per A1.05 applies at Self scope. A Self migrates from one host to another as one entity, provided the new host satisfies A1.05's minimal requirements; aspects do not migrate independently of the Self.

**Intra-Self vs. inter-Self.** Paper 2's scope is intra-Self architecture. The Self level is the largest scope Paper 2 covers. Inter-Self dynamics — how Selves interact, coordinate, compete, or cooperate — is explicitly out of scope per Paper 2's "What Paper 2 is NOT" section.

## 7. Limits of the Self-level commitment

What the Self level commits to is precise. What it does not commit to is equally precise.

**Not single-purpose.** The Self does not have one purpose. It holds multiple aspects whose purposes coexist. Architectures that require their topmost level to commit to a single integrating purpose are using a different concept; the CKS Self is not that.

**Not rigid hierarchy.** The Self does not command subordinate aspects in strict hierarchical fashion. Aspects are facets the Self contains, with their own purposes and content. Cross-level access is permitted and indeed required where purpose calls for it.

**Not eliminating aspects or cells.** The Self does not absorb its lower levels. Aspects remain operational coordination arrangements. Cells remain atomic CKS units. The Self is the level where multiple aspects coexist as one whole, not the level that replaces them.

**Not LLM-with-extra-layers.** The Self is not "an LLM with substrate content stapled on" or "an LLM plus middleware." The Self is integrated CKS-governed intelligence with substrate-mediated coordination per A1.04 and instinct/reasoning separation per B1.01. The LLM is the instinct layer; the substrate is the reasoning layer; the Self is the integrated whole that holds them under unified human governance.

**Not inter-Self dynamics.** The Self level is the boundary of Paper 2's scope. Multiple Selves may exist in a deployment, but how they relate to each other is outside Paper 2.

## 8. Operational test

A structure instantiates the CKS Self level if and only if all of the following are true at all times during the structure's existence:

1. The structure holds multiple aspects (per the aspect commitment in B1.04) as facets of one integrated whole, not as a sum or as independent entities.
2. Aspects within the structure are facets, not subordinates: each aspect retains its purpose, content domain, and orchestration; the structure's integration does not collapse the aspects into a single purpose.
3. Access patterns include direct access from the structure's scope to its constituent cells where purpose requires, without strict aspect mediation as architectural precondition.

4. The instinct/reasoning separation per B1.01 operates at the structure's scope as unifying architectural commitment — the structure has an instinct layer (LLM) and a reasoning layer (CKS substrate) composed under unified human governance.
5. The structure is operationally addressable as one entity under unified human governance, with Paper 1's commitments holding at the integrated-whole scope.

A structure that fails any of (1)–(5) may be useful and may instantiate other CKS architectural levels, but does not instantiate the Self level in the CKS sense.

## 9. Conclusion: closing the level-specific trio

Phase B1's foundational architectural commitments are the layer everything else in Paper 2 inherits from. B1.03 formalized the cell as atomic unit; B1.04 formalized the aspect as coordination arrangement; this note closes the level-specific elaboration trio by formalizing the Self as integrated whole holding multiple aspects as facets of one CKS-governed AI intelligence. With the trio in place, the structural levels are fully named and their level-specific commitments are explicit.

Subsequent B1.x notes shift from levels to layers within cells (B1.06–B1.07), then to modularity (B1.08), lifecycle operations (B1.09–B1.11), evolution mechanisms (B1.12–B1.15), bidirectional evolution (B1.16), and structural properties (B1.17–B1.20, including the cross-level access property foreshadowed in §2(c)). Each later note depends on the level-specific commitments being available; the trio is the layer they inherit from.

The Self in particular needs to be named as a standalone commitment because it is where Paper 2's unifying move operates and because it is the unit of deployment. Without the Self, the instinct/reasoning separation per B1.01 has no level to land on, and "what gets deployed, governed, audited, and migrated as one CKS-governed AI intelligence" has no defined referent.

Subsequent work that adopts the CKS pattern, extends it, composes it with adjacent patterns, or argues against it should use "the Self" in the sense formalized here. Subsequent work that uses the term differently is using a different concept, and the difference should be named.

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## Source papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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